

Module-3

ELECTRODE SYSTEM

Reference electrodes:

The electrode of standard potential, with which we can compare the potentials of another electrode, is called a reference electrode. The best reference electrode used is standard hydrogen electrodes. Its electrode potential at all temperatures is taken as zero.

The different types of electrodes are:

- 1) Metal-Metal ion electrode
- 2) Metal-Metal insoluble electrode
- 3) Gas -ion electrode
- 4) Redox electrode
- 5) Glass electrode

1) Metal-Metal ion electrode: An electrode of this type consists of a metal rod dipped in to a solution of its own salt.

E.g.: Zn rod dipped in a solution of ZnSO_4 (Zn/ZnSO_4)

2) Metal -Metal insoluble electrode: These electrodes consist of a metal in contact with a sparingly soluble salt of the same metal dipped in a solution containing anion of the salt.

E.g.: Calomel electrode ($\text{Hg}/\text{Hg}_2\text{Cl}_2/\text{Cl}^-$)

3) Gas-ion electrode: It consists of an inert metal dipped in a solution containing ions to which the gas is reversible

E.g.: Hydrogen electrode

4) Redox electrode: It consists of an inert metal dipped in a solution containing ions in two oxidation states of the substance.

E.g.: Quinhydrone electrode ($\text{Pt}/\text{Q}/\text{H}_2\text{Q}$)

5) Glass electrode: **E.g.:** $\text{Ag}/\text{AgCl}(\text{s}), \text{HCl} (0.1\text{M})/ \text{Glass}^+$

The calomel electrode potential is depending on the concentration of chloride ions in KCl. If the concentration of chloride ions increases, the potential decreases and vice versa. (The potential of calomel is inversely proportional to the concentration of chloride ions)

The potential of calomel electrode is measured with respect to SHE and it depends on concentration of KCl solution used.

Conc. KCl	0.1N	1N	Saturated KCl
Potential (V)	0.334V	0.281V	0.242V

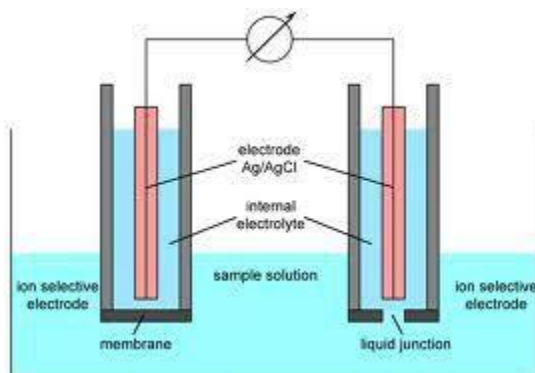
Applications:

- ✓ Used to determine the potential of the other electrodes.
- ✓ It is commonly used as reference electrode in all potentiometric determinations.

Ion selective electrode (ISE):

An ion selective electrode (ISE) is a sensor which converts the activities of a specific ion dissolved in a solution into an electrical potential which can be measured by a voltmeter,

E.g. Glass electrode.

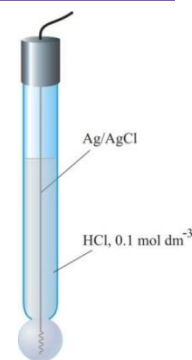
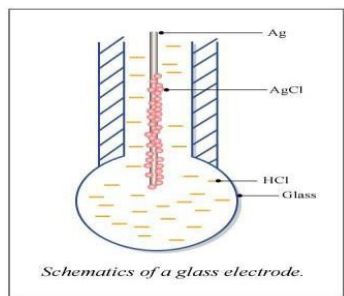


Glass Electrode:

Construction and Working

A glass electrode is a type of ion- selective electrode and consists of a thin- walled glass bulb attached to a glass tube. A very low melting point and high electrical conductivity glass are used for the construction of this bulb. The glass tube contains a dilute solution of constant pH of HCl (0.1N) solution. A silver–silver chloride electrode or platinum wire is immersed as reference electrode in the HCl solution. The working of glass electrode is based upon the observation that when a glass surface is in contact with a solution, there exists a potential difference between the glass surface and the solution, the magnitude of which depends upon the H^+ ion concentration of the solution and the nature of glass. The glass electrode may be represented as
 $Ag, AgCl(s) / 0.1 N HCl / glass / H^+ = unknown$

The electrode potential of the glass electrode depends up on the concentration of H^+ ions contained in the experimental solution and are given by.



$$E_G = E_G^0 + \frac{2.303RT}{F} \log H^+$$

$$= E_G^0 - 0.0591 \log H^+$$

$$= E_G^0 + 0.0591 \text{pH}$$

E_G^0 is the standard electrode potential, i.e. the potential of the glass electrode when the solution contains unit concentration of H^+ ions. The value of E_G^0 depends on the nature of the glass used in the construction of the bulb.

The pH of the solution can be determined if the potential of the glass electrode is known. To determine the value of glass electrode potential (E_G), the glass electrode is combined with a reference electrode such as calomel electrode. The EMF of this cell will be given by

$$E_{\text{cell}} = E_{\text{(calomel)}} - E_{\text{(glass)}}$$

$$= E_G^0 - E_G - 0.0591 \text{pH}$$

$$\text{pH} = \frac{E_G - E_G^0 - E_{\text{cell}}}{0.0591}$$

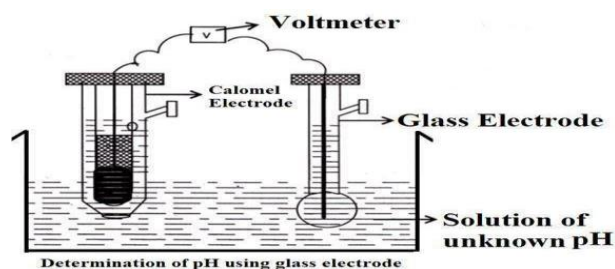
Advantages of glass electrode:

1. Glass electrode is easy to operate and equilibrium is rapidly reached.
2. The pH value obtained is very accurate.

Limitations of glass electrode:

1. Glass electrode can be used in solutions with pH range 0-10, because electrodes are composed of special glass that can be used up to pH 12.
2. The resistance is extremely high in the order of 10 to 100 million ohms, which cannot be measured by ordinary potentiometers and special electronic potentiometers have to be used.

Determination of pH using glass electrode:



- For pH determination Glass electrode is immersed in the solution whose pH is to be determined and it is combined with a reference electrode such as a calomel electrode(saturated) through a salt bridge.
- The cell assembly is represented as, $\text{Hg} | \text{Hg}_2\text{Cl}_2 | \text{KCl} || \text{Solution of unknown pH} | \text{glass} | 0.1\text{M HCl} | \text{Ag}^+ | \text{AgCl}$

Step 1:

The emf of the above cell, E_{cell} is measured using an electronic **voltmeter or pH meter**.

- The emf of the cell is given by:

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = E_{\text{glass}} - E_{\text{cal}}$$

Since E_{cal} , E_{cell} is known, so E_{glass} can be evaluated.

The potential of glass electrode E_G is given by, $E_G = L^1 - 0.0591\text{pH} E_{\text{cell}} = E_G - E_{\text{cal}}$

$$\text{pH} = \frac{L^1 - E_{\text{cell}} - E_{\text{cal}}}{0.0591}$$

$$E_{\text{cell}} = L^1 - 0.0591\text{pH} - E_{\text{cal}}$$

Step 2:

L^1 value is evaluated by dipping the glass electrode in a solution of known pH and measuring the emf of the cell formed when combined with a calomel.

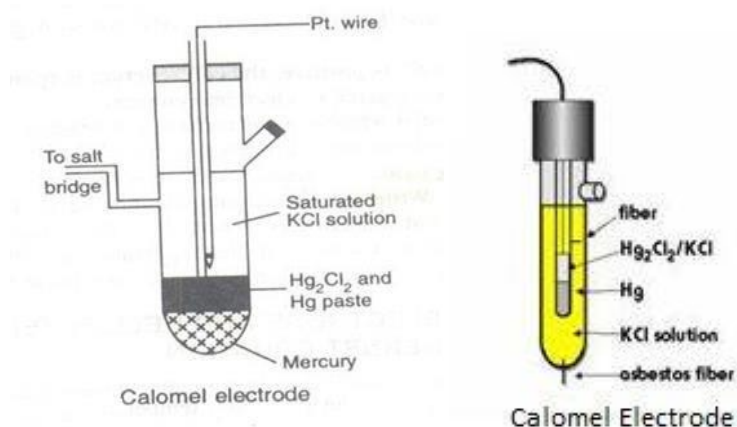
$$E_{\text{Cell}} = E_G - E_{\text{cal}}$$

$$E_{\text{Cell}} = L' - 0.0591 \text{ pH}_{(\text{known})} - E_{\text{cal}}$$

$$L' = E_{\text{Cell}} + 0.0591 \text{ pH}_{(\text{known})} + E_{\text{cal}}$$

Saturated Calomel Electrode:

Construction and Working



Calomel electrode is a metal-metal salt ion electrode. It consists of mercury, mercurous chloride and a solution of KCl. Mercury is placed at the bottom of a glass tube having a side tube on each side. Mercury is covered by a paste of mercurous chloride (calomel) with mercury & KCl.

A solution of KCl is introduced above the paste through the side tube. A platinum wire sealed in a glass tube is dipped into mercury and used to provide the external electrical contact. The concentration of KCl used is either decinormal, normal (or) saturated. The electrode is known as decinormal, normal, saturated calomel electrode.

The electrode whose potential is to be determined is connected to this electrode through salt bridge. The potential of electrode depends upon the

concentration of KCl solution.

The net reversible electrode reaction is;



NERNST EQUATION:

$$E = E^0 - \frac{2303RT}{\text{Hg}_2\text{Cl}_2 \cdot 2F} \log[\text{Cl}^-]^2$$

$$= E^0 - \frac{2.303RT}{F} \log [\text{Cl}^-]$$

$$= E^0 - 0.0591 \log[\text{Cl}^-]$$

The electrode potential depends up on the conc. of KCl solution (i.e. chloride ions) At 25°C,

For Saturated KCl solution electrode potential is 0.2415 volts.

For 1N KCl solution standard reduction potential is 0.281 volts. For 0.1N KCl solution the reduction potential is 0.3338 volts.

Applications:

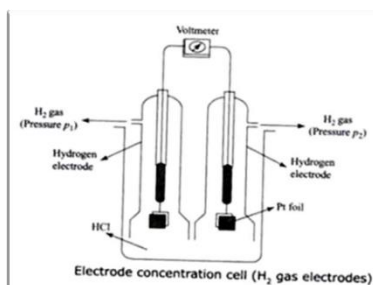
1. It is used as a secondary reference electrode in the measurement of single electrode potential.
2. It is the most commonly used reference electrode in all potentiometric determinations.

Concentration cells:

In the concentration cell, the EMF is produced due to the difference in the concentration of the electrodes or in the concentration of electrolyte, i.e., a concentration cell is an electrochemical cell which produces electrical energy by the transfer of a material from a system at higher concentration to a system at lower concentration. The difference in concentration may be due to the difference in concentration of electrodes or electrolyte. Based on this concentration, cells are classified into groups:

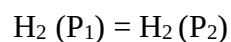
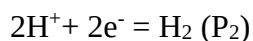
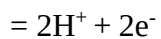
CONSTRUCTION

In these cells, the potential difference is developed between two like electrodes at different concentrations dipped in the same solution of the electrolyte. For example, two hydrogen electrodes at different gas pressure dipped in the same solution of hydrogen ions constitute a cell of this type.



Pt/H₂ (pressure P₁) / H⁺ (a) / (H₂ (pressure P₂) / Pt

If P₁ > P₂ oxidation occurs at L.H.S. electrode and reduction occurs at R.H.S electrode H₂ (P₁)



$$E_{\text{cell}} = \frac{0.0591}{2} \log \frac{P_1}{P_2}$$